

END SEMESTER EXAMINATION 2024

Name of the Program: B.Tech. (EE).

Semester: 1st

Name of the Course: Basic Electrical Engineering

Course Code: TEE 101

Maximum Marks: 100

Time: 3 hrs

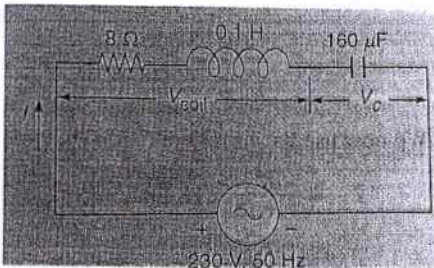
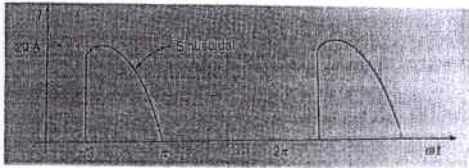
Note

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

1		20 Marks
(a)	Define the following terms with suitable examples: (i) Frequency and Time Period (ii) Open Circuit and Short circuit condition (iii) Supermesh condition (iv) Active power, reactive power, and apparent power (v) Form factor and Peak factor	CO1
(b)	A coil of resistance 8Ω and inductance 0.1H is connected in series with a condenser $160\mu\text{F}$ capacitance across a 230V, 50Hz supply as shown below. Calculate (i) Inductive reactance (ii) the capacitive reactance (iii) the circuit impedance, (iv) current (v) power factor, and (vi) voltage of coil (V_{coil}) and condenser (V_c) respectively.  Fig: 1	CO1
(c)	Controlled rectifiers are employed to convert ac to dc. The output of controlled rectifier has wave form as shown below. Find average value, rms value, form factor and peak factor.  Fig: 2	CO1
2		20 Marks
(a)	Using nodal analysis method find V_x for the circuit given below. Describe Kirchhoff's current law.	CO2

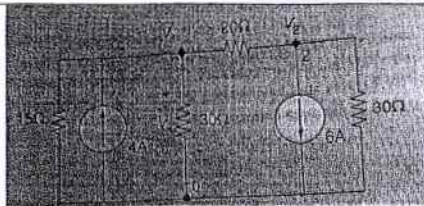


Fig: 3

- (b) State and explain Norton's Theorem. Find the Norton equivalent across X-Y

CO2

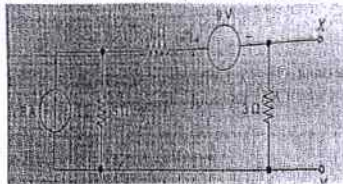


Fig: 4

- (c) Define voltage triangle, impedance triangle, and power triangle for single phase series R-C circuit.

CO2

3

20 Marks

- (a) State superposition theorem and Find the current in 5Ω using the principle of superposition theorem.

CO3



Fig: 5

- (b) Find the Thevenin equivalent as viewed by resistance R. Find the value of R for maximum power dissipation in it and the value of this power.

CO3



Fig: 6

- (c) Voltage and current for a circuit with two elements in series are expressed as follows:

CO3

$$v(t) = 170 \sin(6280t + \frac{\pi}{3})$$

$$i(t) = 8.5 \sin(6280t + \frac{\pi}{2})$$

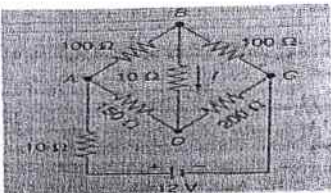
- Plot the two waveforms
- Determine the frequency in Hz
- Determine the power factor stating its nature
- What are the values of the elements?

4

20 Marks

- (a) Define the term "Electrical Cable". Give classification of electrical cable. Explain any two in detail.

CO4

(b)	What do you understand by Circuit Breakers? Give classification of circuit breakers. What is the difference between miniature circuit breaker (MCB) and moulded-miniature circuit breaker (MCCB)?	CO4
(c)	Three phasors are $A=10+j10$ (rectangular form); $B=50\angle 120^\circ$ (polar form) and $C=8-j6$ (rectangular form). Evaluate $\frac{AB}{C}$ and express it in rectangular and polar form.	CO4
5		20 Marks
(a)	What do you understand by "Electrical Grounding" in an electrical system? What are the basic needs of electrical earthing (grounding)? Also define different types of electrical earthing?	CO5
(b)	Define Kirchoff's Voltage law. Find the current through 10Ω resistor connected between terminals B-D using mesh analysis.	CO5
 Fig. 7		
(c)	The instantaneous values of two alternating voltages are represented by $v_1=60\sin\theta$ and $v_2=40\sin(\theta - \frac{\pi}{3})$. Derive expression for the instantaneous values of (i) the sum and (ii) the difference, of these voltages.	CO5

